STATISTICS

The Only 10 Statistics Questions Asked in Every Data Interview

The 10 statistics questions that come up again and again in data analyst and data scientist interviews, each with a clear answer and an example you can say out loud.

INSIDE THIS PDF · 8 PAGES

- Questions 1 and 2
- Questions 3 and 4
- Questions 5 and 6
- Questions 7 and 8
- Question 9: A/B test design
- Question 10: Simpson's paradox
- How to practise these

@thedataguy16

Questions 1 and 2

The warm-up questions. Get these smooth.

1. Mean vs median: when do you use which?

Median when data is skewed or has outliers, like salaries or order values. Mean when data is roughly symmetric.

Example to say

"Average order value was ₹2,400 but median was ₹900. A few bulk orders pulled the mean up, so I reported the median."

2. What is standard deviation?

How spread out values are around the average, in the same units as the data.

Example to say

"Two stores both average ₹1 lakh a day, but one has SD ₹5,000 and the other ₹40,000. The second is far less predictable."

@thedataguy16

Questions 3 and 4

The most asked statistics questions overall.

3. Explain p-value simply

If nothing really changed, the probability of seeing a result at least this extreme. Small p-value means the result is unlikely to be just chance.

Trap to avoid

It is not "the probability the null hypothesis is true".

4. Type I vs Type II error

Type I: false positive, you say there is an effect when there is none. Type II: false negative, you miss a real effect.

Example to say

"Type I: launching a new checkout that does not actually help. Type II: dropping one that would have helped because the test was too small."

@thedataguy16

Questions 5 and 6

Show you understand samples and uncertainty.

5. What is the Central Limit Theorem?

Averages of large enough samples are roughly normally distributed, whatever the shape of the original data.

Why it matters

It is why t-tests and confidence intervals work on skewed business data like revenue.

6. What is a confidence interval?

A range that would contain the true value in 95% of repeated samples.

Example to say

"Conversion lift is 2.1%, with a 95% confidence interval of 0.8% to 3.4%. The whole range is above zero, so the lift is real."

@thedataguy16

Questions 7 and 8

Applied questions. Interviewers want the business link.

7. Correlation vs causation

Correlation means two things move together. Causation means one causes the other. A third factor can drive both.

Example to say

"Customers who use the app more spend more. But high spenders may just like the app. Only an experiment proves the app causes spending."

8. How do you detect outliers?

IQR rule: below Q1 − 1.5×IQR or above Q3 + 1.5×IQR. Or z-score above 3. Then decide: error, or a real extreme case?

Key point

Never delete outliers without checking why they exist.

@thedataguy16

Question 9: A/B test design

"How would you test a new checkout button?" Answer in this order.

1. Metric

One primary metric: checkout conversion rate. Plus guardrails like revenue per user.

2. Hypothesis

H0: no difference. H1: new button changes conversion.

3. Sample size

Based on current rate, smallest lift worth detecting, 5% significance and 80% power.

4. Run

Random 50/50 split by user, full weeks, no stopping early.

5. Decide

Check significance and confidence interval, then guardrails, then ship or not.

@thedataguy16

Question 10: Simpson's paradox

Asked to see if you check results by segment.

What it is

A trend that appears in every group reverses when the groups are combined.

Example

Version B converts better on mobile and on desktop separately, but worse overall, because B got far more mobile traffic, which converts lower.

What to say

"I always break results by key segments like device and new vs returning users before concluding."

Pattern across all 10: definition in one line, then a business example. That combination is what scores.

@thedataguy16

How to practise these

Two days is enough if you do it this way.

- **Day 1**

  Write your own one-line answer and one example for each question, from your work or projects.

- **Day 2**

  Say all 10 out loud, under 60 seconds each. Record and listen back.

- **Before the interview**

  Re-read questions 3, 4 and 9. They come up most.

@thedataguy16